

HTML Tags & HTML Controls

A Complete Beginner's Guide

Learning Objectives

By the end of this lesson, you will be able to:

- Understand what HTML tags are.
 - Create a basic HTML webpage.
 - Use various text formatting tags.
 - Display mathematical and chemical expressions.
 - Display preformatted text and source code.
 - Organize webpages using headings and paragraphs.
 - Use address, blockquote, marquee, links and meta tags.
 - Create hyperlinks using anchor tags.
 - Display images.
 - Embed webpages using iframes.
 - Create ordered, unordered and description lists.
-

What is HTML?

HTML (HyperText Markup Language) is the standard language used to create web pages.

HTML is **not** a programming language.

It is a **markup language**, meaning it tells the browser:

- what content exists
- where it should appear
- what type of content it is

Example:

```
This is a heading.  
This is a paragraph.
```

```
this is a paragraph.  
This is an image.  
This is a button.
```

HTML provides tags to describe each of these.

What is an HTML Tag?

An HTML tag tells the browser how a particular piece of content should be displayed.

Example

```
<p>Hello World</p>
```

Here,

- `<p>` → Opening Tag
 - `Hello World` → Content
 - `</p>` → Closing Tag
-

Structure of an HTML Element

```
<opening tag>  
  Content  
</closing tag>
```

Example

```
<h1>Welcome</h1>
```

Basic HTML Boilerplate

Every HTML page begins with a basic structure.

```
<!DOCTYPE html>  
<html lang="en">  
  
<head>
```

```
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>My First Webpage</title>
</head>

<body>

  <h1>Welcome to HTML</h1>
  <p>This is my first webpage.</p>

</body>

</html>
```

Explanation of Boilerplate

<!DOCTYPE html>

Tells the browser this document uses HTML5.

<html>

Root element of the webpage.

Everything is written inside it.

<head>

Contains information about the webpage.

Examples:

- title
 - meta information
 - CSS
 - JavaScript
-

<meta charset="UTF-8">

Allows special characters and multiple languages.

<meta name="viewport"...>

Makes the webpage responsive on mobile devices.

<title>

Appears on the browser tab.

<body>

Contains everything visible on the webpage.

1. Text Formatting Tags

These tags change the appearance or meaning of text.

** — Bold Text**

Displays text in bold.

```
<p>This is <b>bold</b> text.</p>
```

Output

```
This is bold text.
```

** — Important Text**

Makes text bold and also indicates that it is important.

```
<p>This is <strong>important</strong>.</p>
```

Difference:

```
<b>
```

- Only makes text bold.

```
<strong>
```

- Makes text bold
- Gives semantic meaning (important).

<i> — Italic Text

```
<p>This is <i>italic</i> text.</p>
```

** — Emphasized Text**

```
<p>Please <em>read carefully</em>.</p>
```

Difference:

```
<i>
```

- Only changes appearance.

```
<em>
```

- Adds emphasis.

<u> — Underlined Text

```
<p>This is <u>underlined</u>.</p>
```

Complete Example

```
<!DOCTYPE html>
<html>

<head>
  <title>Formatting Tags</title>
</head>
```

```
<body>

  <h2>Formatting Tags</h2>

  <p><b>Bold Text</b></p>
  <p><strong>Important Text</strong></p>
  <p><i>Italic Text</i></p>
  <p><em>Emphasized Text</em></p>
  <p><u>Underlined Text</u></p>

</body>

</html>
```

2. Superscript and Subscript

Superscript `<sup>`

Displays text above the normal line.

Used for:

- powers
- exponents

Example

```
<p>x<sup>2</sup></p>
```

Output

x²

More Examples

```
<p>a<sup>3</sup></p>
```

```
<p>10<sup>5</sup></p>
```

Subscript <sub>

Displays text below the normal line.

Used in:

- Chemistry
- Mathematics

Example

```
<p>H<sub>2</sub>O</p>
```

Output

```
H2O
```

More Examples

```
<p>CO<sub>2</sub></p>
```

```
<p>H<sub>2</sub>SO<sub>4</sub></p>
```

```
<p>NH<sub>3</sub></p>
```

Complete Example

```
<!DOCTYPE html>
<html>

<head>
  <title>Superscript and Subscript</title>
</head>

<body>

  <h2>Mathematical Expressions</h2>

  <p>x<sup>2</sup></p>
  <p>a<sup>3</sup> + b<sup>3</sup></p>
```

```
<h2>Chemical Formulas</h2>
```

```
<p>H<sub>2</sub>O</p>
```

```
<p>CO<sub>2</sub></p>
```

```
<p>H<sub>2</sub>SO<sub>4</sub></p>
```

```
</body>
```

```
</html>
```

3. Preformatted Text and Code

<pre>

Preserves:

- spaces
- tabs
- line breaks

Exactly as written.

Example

```
<pre>
Name : John
Age  : 20
City : Rajkot
</pre>
```

Output

```
Name : John
Age  : 20
City : Rajkot
```

<code>

Used to display programming code.

Example

Example

```
<code>
int a = 10;
</code>
```

<pre> with <code>

```
<pre>
<code>
#include <stdio.h>

int main()
{
    printf("Hello World");
    return 0;
}
</code>
</pre>
```

Complete Example

```
<!DOCTYPE html>
<html>

<head>
    <title>Pre and Code</title>
</head>

<body>

<h2>Preformatted Text</h2>

<pre>
Student Name : Rahul
Course       : BCA
Semester     : 3
</pre>

<h2>Code Example</h2>

<pre>
<code>
int a = 10;
int b = 20;

printf("%d\n", a + b);
```

```
    print(' %d , %d' % (a+b),  
</code>  
</pre>  
  
</body>  
  
</html>
```

4. Heading Tags and Paragraph

HTML provides six heading levels.

```
<h1>  
<h2>  
<h3>  
<h4>  
<h5>  
<h6>
```

Largest



Smallest

Example

```
<h1>Main Heading</h1>  
  
<h2>Sub Heading</h2>  
  
<h3>Section</h3>  
  
<h4>Topic</h4>  
  
<h5>Sub Topic</h5>  
  
<h6>Small Heading</h6>
```

Paragraph

```
<p>  
HTML is the standard markup language for creating webpages
```

HTML is the standard markup language for creating webpages.

Complete Example

```
<!DOCTYPE html>
<html>

<head>
<title>Headings</title>
</head>

<body>

<h1>Heading 1</h1>

<h2>Heading 2</h2>

<h3>Heading 3</h3>

<h4>Heading 4</h4>

<h5>Heading 5</h5>

<h6>Heading 6</h6>

<p>
This is a paragraph explaining HTML.
</p>

</body>

</html>
```

5. Other Important Tags

<address>

Displays contact information.

```
<address>

ABC College<br>
```

```
Rajkot<br>
Gujarat<br>
Email : info@abc.com
</address>
```

<blockquote>

Displays a long quotation.

```
<blockquote>
Education is the passport to the future.
</blockquote>
```

<marquee>

Creates scrolling text.

```
<marquee>
Welcome to HTML Class
</marquee>
```

Link (<link>)

Used inside the <head> section to connect external resources such as CSS.

```
<head>
  <link rel="stylesheet" href="style.css">
</head>
```

<meta>

Provides metadata about the webpage.

Example

```
<meta charset="UTF-8">  
  
<meta name="viewport"  
content="width=device-width, initial-scale=1.0">
```

6. Anchor Tag and Image

Anchor Tag

Creates hyperlinks.

Syntax

```
<a href="URL">Link Text</a>
```

Example

```
<a href="https://www.google.com">  
Visit Google  
</a>
```

Open in new tab

```
<a href="https://www.google.com"  
target="_blank">  
Google  
</a>
```

Image Tag

Displays images.

Syntax

```

```

Example

```

```

Complete Example

```
<!DOCTYPE html>
<html>

<head>
<title>Anchor and Image</title>
</head>

<body>

<a href="https://www.google.com"
target="_blank">

Visit Google

</a>

<br><br>


```

</html>

7. Iframe Tag

An iframe embeds another webpage inside the current webpage.

Syntax

```
<iframe  
src="https://www.wikipedia.org"  
width="600"  
height="400">  
</iframe>
```

Example

```
<!DOCTYPE html>
<html>

<head>
<title>Iframe Example</title>
</head>

<body>

<h2>Embedded Website</h2>

<iframe
src="https://example.com"
width="600"
height="300">
</iframe>

</body>

</html>
```

8. List Controls

HTML supports three types of lists.

Unordered List

Uses bullets.

```
<ul>

<li>Apple</li>

<li>Mango</li>

<li>Orange</li>

</ul>
```

Output

- Apple
- Mango
- Orange

Ordered List

Uses numbers.

```
<ol>

<li>HTML</li>

<li>CSS</li>

<li>JavaScript</li>

</ol>
```

Output

1. HTML

2. CSS
3. JavaScript

Ordered List (`<ol>`) - `start` Attribute

The `start` attribute is used with the `<ol>` (Ordered List) tag to specify the number or position from which the list should begin.

The `start` attribute works with all numbering styles:

- Numbers (`1`)
- Uppercase Letters (`A`)
- Lowercase Letters (`a`)
- Uppercase Roman Numerals (`I`)
- Lowercase Roman Numerals (`i`)

Note: The `start` attribute **always accepts a numeric value**, even for alphabetic and Roman numeral lists. The browser automatically converts the number to the appropriate letter or Roman numeral based on the `type` attribute.

1. Numbered List (Default)

Starts numbering from the specified number.

```
<ol start="5">
  <li>HTML</li>
  <li>CSS</li>
  <li>JavaScript</li>
</ol>
```

Output

- ```
5. HTML
6. CSS
7. JavaScript
```

---

## 2. Uppercase Alphabet ( `type="A"` )

---

The browser converts the numeric value into the corresponding uppercase letter.

## Example 1: Start from C

---

```
<ol type="A" start="3">
 Apple
 Banana
 Mango

```

### Output

```
C. Apple
D. Banana
E. Mango
```

---

## Example 2: Start from H

---

```
<ol type="A" start="8">
 HTML
 CSS
 JavaScript

```

### Output

```
H. HTML
I. CSS
J. JavaScript
```

---

# 3. Lowercase Alphabet ( type="a" )

---

The browser converts the numeric value into the corresponding lowercase letter.

## Example 1: Start from c

---

```
<ol type="a" start="3">
```

```
Dog
Cat
Cow

```

## Output

```
c. Dog
d. Cat
e. Cow
```

---

## Example 2: Start from m

```
<ol type="a" start="13">
 January
 February
 March

```

## Output

```
m. January
n. February
o. March
```

---

# 4. Uppercase Roman Numerals ( `type="I"` )

The browser converts the numeric value into uppercase Roman numerals.

---

## Example 1: Start from V

```
<ol type="I" start="5">
 Chapter One
 Chapter Two
 Chapter Three

```

## Output

V. Chapter One  
VI. Chapter Two  
VII. Chapter Three

---

## Example 2: Start from X

---

```
<ol type="I" start="10">
 Topic A
 Topic B
 Topic C

```

### Output

X. Topic A  
XI. Topic B  
XII. Topic C

---

## 5. Lowercase Roman Numerals ( `type="i"` )

---

The browser converts the numeric value into lowercase Roman numerals.

### Example 1: Start from v

---

```
<ol type="i" start="5">
 Section One
 Section Two
 Section Three

```

### Output

v. Section One  
vi. Section Two  
vii. Section Three

---

### Example 2: Start from v

## Example 2. Start from x

---

```
<ol type="i" start="10">
 Introduction
 Body
 Conclusion

```

## Output

```
x. Introduction
xi. Body
xii. Conclusion
```

---

# Complete Example

---

```
<!DOCTYPE html>
<html>

<head>
 <title>Ordered List – Start Attribute</title>
</head>

<body>

 <h2>1. Numbered List</h2>
 <ol start="5">
 HTML
 CSS
 JavaScript

 <h2>2. Uppercase Alphabet</h2>
 <ol type="A" start="3">
 Apple
 Banana
 Mango

 <h2>3. Lowercase Alphabet</h2>
 <ol type="a" start="3">
 Dog
 Cat
 Cow

 <h2>4. Uppercase Roman Numerals</h2>
 <ol type="I" start="5">
```

```
<ol type="I" start="5">
 Chapter One
 Chapter Two
 Chapter Three

<h2>5. Lowercase Roman Numerals</h2>
<ol type="i" start="5">
 Section One
 Section Two
 Section Three

</body>

</html>
```

---

## Summary Table

---

type Attribute	start Value	First Item Displayed
(default)	5	5
A	3	C
A	8	H
a	3	c
a	13	m
I	5	V
I	10	X
i	5	v
i	10	x

---

## Key Points

---

- The `start` attribute specifies the starting position of an ordered list.
  - It **always accepts a numeric value**, regardless of the numbering style.
  - The browser automatically converts the numeric value into:
    - Numbers ( `1`, `2`, `3...` )
    - Uppercase letters ( `A`, `B`, `C...` )
    - Lowercase letters ( `a`, `b`, `c...` )
-

- Uppercase Roman numerals ( I, II, III... )
- Lowercase Roman numerals ( i, ii, iii... )
- The `start` attribute can be combined with the `type` attribute to create customized ordered lists.

---

## Description List ( `<dl>`, `<dt>`, `<dd>` )

---

A **Description List** is used to display a list of **terms and their descriptions (definitions)**.

Unlike ordered ( `<ol>` ) and unordered ( `<ul>` ) lists, a description list does **not** use numbers or bullets.

A description list consists of three tags:

- `<dl>` - Description List
- `<dt>` - Description Term
- `<dd>` - Description Details (Definition)

---

## Complete Example

---

```
<!DOCTYPE html>
<html>

<head>
<title>Lists</title>
</head>

<body>

<h2>Ordered List</h2>

HTML
CSS
JavaScript

<h2>Unordered List</h2>

Apple
Mango
Orange

<h2>Description List</h2>
```

```
</h2>Description List</h2>
```

```
<dl>
```

```
<dt>HTML</dt>
```

```
<dd>HyperText Markup Language</dd>
```

```
<dt>CSS</dt>
```

```
<dd>Cascading Style Sheets</dd>
```

```
</dl>
```

```
</body>
```

```
</html>
```

---

## Summary

---

In this lesson, you learned:

- HTML boilerplate structure
- Text formatting tags ( `<b>` , `<strong>` , `<i>` , `<em>` , `<u>` )
- Superscript ( `<sup>` ) and subscript ( `<sub>` )
- Preformatted text ( `<pre>` ) and code ( `<code>` )
- Heading tags ( `<h1>` to `<h6>` ) and paragraphs ( `<p>` )
- Address, blockquote, marquee, link, and meta tags
- Anchor ( `<a>` ) tag for hyperlinks
- Image ( `<img>` ) tag for displaying images
- Iframe ( `<iframe>` ) for embedding webpages
- Ordered, unordered, and description lists

These HTML tags form the foundation of webpage structure and content formatting. Mastering them is essential before moving on to CSS and JavaScript.